

MechanicaFluidorum

A Verifier-in-the-Loop Program on 3D Navier–Stokes Regularity

Dual-Scale Geometry, the Symmetric-Square Lock, and Hypothesis U

Scientific Progress Report — Intermediate Results

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Abstract

This report documents the state of a staged, verifier-in-the-loop research programme on the global regularity of the three-dimensional incompressible Navier–Stokes equations (NSE), organised around *Hypothesis U* — a uniform-in-cutoff enstrophy bound for the frequency-truncated system. The programme’s distinguishing feature is not a claimed proof but a *method*: every assertion carries an explicit epistemic tier, mechanically enforced by a two-gate verification pipeline (exact rational arithmetic; Lean 4 kernel compilation with a pinned axiom footprint). We report 66 kernel-verified theorems across seven Lean files, seven exact-arithmetic certification harnesses, and — equally importantly — a documented record of what has *failed*: two abandoned numerical designs, five caught errata (one in a formula sourced from the literature, one in the programme’s own design memo, one a broken commit, one a false positive in the verification gate itself, and one an unpinned dependency that would have verified the programme against the wrong prover library), and four analytical tracks that remain correctly blocked on undefined objects. We state precisely what is established, what is merely checkable, and what remains conjecture. **No claim is made that Hypothesis U is true, nor that the Navier–Stokes problem is resolved.** The Millennium Prize problem remains open, and this programme’s central hypothesis is a restatement of its open core, not a reduction of its difficulty.

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1 Context Management: Reading This Document and This Repository

Purpose of this section

This programme is executed collaboratively by a human mathematician and AI agents across many sessions. Context is therefore a *managed artifact*, not an accident of memory. This section specifies where authoritative context lives, how it is kept honest, and what a reader (human or machine) must load before acting. It is placed first deliberately: acting on this programme without this context has already caused documented failures (§10).

1.1 The governing documents and their precedence

Context is deliberately *not* held in any individual's or agent's memory. It is externalised into five documents with a strict precedence order:

Document	Role	Authority
SPEC.md	Normative rulebook	Highest. Defines the tier system, the honesty clause, the obstruction ledger. Where any other document conflicts with it, SPEC.md wins.
PLAN.md	Operational law	Task list with Definition-of-Done criteria and the escalation protocol (E-1...E-5). Wins over calendar/roadmap.
LEDGER.md	Claim inventory	Normative: a claim not listed here has no tier and may not be cited.
LL.md	Lessons learned	Why each process rule exists, with the incident that produced it.
CLAUDE.md	Build/verify manual	Commands, architecture, Mathlib location, working conventions.

1.2 The epistemic charter

The organising principle, stated in SPEC.md §0, is a division of labour:

The machine verifier checks the proofs; the human mathematician audits the questions.

Machine verification establishes that a proof is valid. It cannot establish that a *statement* means what the mathematics intends — that a formalisation is *adequate* to its informal gloss. Adequacy is reserved to human audit. Consequently **no LLM output gates a tier promotion**, and several artifacts in this repository are compiled, kernel-clean, and nonetheless explicitly marked DRAFT pending human statement-adequacy audit (§6).

1.3 The three-tier gating system

Tier	Meaning	Mechanically enforced gate
Tier C	Conjecture, analogy, physical narrative, unverified reduction	None, but must be tagged. Confined to <code>exploration/</code> or <code>docs/narrative/</code> . Floating-point code lives here and <i>only</i> here.
Tier B	Checkable: identities validated in exact rational arithmetic	Harness exits 0. <code>fractions.Fraction</code> or <code>int</code> only — floats are banned. Every harness ships a negative control demonstrated to actually fail.
Tier A	Established: Lean 4 kernel-compiled	Zero <code>sorry</code> ; axiom footprint naming <i>no axiom outside</i> <code>{propext, Classical.choice, Quot.sound}</code> . A strict subset is cleaner and passes — see §10, where the earlier literal-equality reading is recorded as a defect.

Two subtleties are load-bearing and were learned the hard way:

1. **“No `sorry` in the source” is not the Tier A gate.** A failed proof still defines its theorem name in the environment; only `#print axioms` reveals a stray `sorryAx`. The gate is the *axiom footprint*, checked on the compiled artifact.
2. **A checker that cannot fail is not a checker.** Every Tier B harness must contain a negative control that is *demonstrated* to fail when the perturbation is applied. This report documents cases where an intended negative control turned out to be undetectable by construction, and had to be redesigned (§9).

1.4 The honesty clause and the obstruction ledger

Every public claim carries its tier. Tier A status attaches to the *formal statement actually proven*, never to its informal gloss. Additionally, `SPEC.md` §1.3 requires every proposed strategy — human or agent-generated — to file an *Obstruction Compliance Note* addressing five known barriers (Table 1). Strategies that cannot are rejected at review, before any formal work begins.

ID	Obstruction	Compliance question the strategy must answer
O1	Tao’s averaged NSE blow-up	Which structural feature of the <i>true</i> nonlinearity, beyond the energy identity, does the strategy use?
O2	Supercriticality of enstrophy	Where exactly does the strategy gain over the naive enstrophy inequality, <i>uniformly</i> in α' ?
O3	Caffarelli–Kohn–Nirenberg	If the strategy implies something stronger than CKN, what new mechanism pays for it?
O4	Buckmaster–Vicol non-uniqueness	In which class is the $\alpha' \rightarrow 0$ limit taken, and why is the limit the physical solution?
O5	The Euler test	Point to the step that fails at $\nu = 0$. A strategy that would also “prove” 3D Euler regularity is presumptively wrong.

Table 1: The obstruction ledger (`SPEC.md` §1.3). These are the barriers any credible attack must address; they are stated here because they constrain everything in §7.

1.5 Reproducing the verification state

The only thing that counts as “it works” is the verification pipeline:

```
./scripts/verify.sh  # Gate 1 : Tier B exact-arithmetic harnesses
                    # Gate 1b : LEDGER consistency + single-active-file lint
                    # Gate 2 : Lean kernel compile + axiom-footprint membership test
```

Every dataset in `data/` ships a `.meta` sidecar recording the exact generating command, tool versions, and `sha256sum`, so results can be audited rather than merely re-read.

Gate 1b (added 2026-08-13). `LEDGER.md` is normative — a claim not listed there may not be cited — so an *orphan* row, one citing a theorem or file that no longer exists, silently degrades the whole inventory. Gate 1b checks that every repo path cited in the ledger exists, that every Lean name in a Tier A row is actually declared, and that no banned versioned duplicates (`_v2`, `_final`) have appeared. It carries its own negative controls, run on every invocation: a dangling path, a dangling name, and a banned filename must each be *rejected*, and a clean case must be *accepted*. It found one real defect on its first run (a row citing the non-identifier fragment `_half` rather than `step1_flux_bound_half`).

A latent reproducibility defect, found by doing the cold build

`lean_src/` was made a standalone Lake project on 2026-08-13, so Gate 2 no longer depends on an external Mathlib checkout this repository does not control. (That coupling was not hypothetical: an unrelated build running in that shared checkout rebuilt its `.olean` files mid-run and failed Gate 2 on a file this repository had not touched.)

Doing the build surfaced something more serious than the inconvenience that motivated it. The project's `lakefile` declared its dependency as `require mathlib from git with no revision pinned`. A cold build from that specification resolves to Mathlib *master* — which in general does not build against the pinned toolchain, and, worse, would have silently verified every result against a *different Mathlib* than the one all existing `LEDGER.md` claims were checked with. For a programme whose entire epistemic claim rests on machine-checkability, an unpinned verifier is a defect in the foundation, not in the build script. Mathlib is now pinned by revision, and the lockfile pinning every transitive dependency is tracked rather than ignored.

2 The Problem, Stated Honestly

2.1 Classical setting

The incompressible Navier–Stokes equations on the torus $\mathbb{T}^3$ are

$$\partial_t u + (u \cdot \nabla)u = -\nabla p + \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(0) = u_0. \quad (1)$$

Leray established global existence of weak solutions [1]. The Clay Millennium Prize problem, as formalised by Fefferman [2], asks for a proof of *exactly one* of four statements — (A) existence and smoothness on $\mathbb{R}^3$; (B) existence and smoothness on $\mathbb{R}^3/\mathbb{Z}^3$ (the periodic case, which is the setting of this programme); (C) breakdown on $\mathbb{R}^3$; (D) breakdown on $\mathbb{R}^3/\mathbb{Z}^3$. Note that a *breakdown* result resolves the problem just as a regularity result does; this is consistent with the present programme's stance that a negative verdict is a successful outcome. The obstruction is *supercriticality*: the natural energy bound is too weak to control the nonlinearity at small scales.

2.2 The regularised family and Hypothesis U

For $\alpha' > 0$ let $J_{\sqrt{\alpha'}}$ denote the projection onto Fourier modes $|k| \leq 1/\sqrt{\alpha'}$, and let $u^{(\alpha')}$ solve the truncated system

$$\partial_t u + ((J_{\sqrt{\alpha'}} u) \cdot \nabla)u = -\nabla p + \nu \Delta u, \quad \nabla \cdot u = 0, \quad u(0) = J_{\sqrt{\alpha'}} u_0, \quad (2)$$

which is globally smooth for each fixed $\alpha' > 0$.

Hypothesis 1 (Hypothesis U). For data u_0 and horizon T ,

$$\sup_{\alpha' > 0} \sup_{0 \leq t \leq T} \|\nabla u^{(\alpha')}(t)\|_{L^2(\mathbb{T}^3)} < \infty. \quad (3)$$

Proposition 2.1 (Millennium Reformulation; **Tier C** at paper level). *If Hypothesis U holds for all smooth divergence-free u_0 and all T , then 3D Navier–Stokes is globally regular. The chain is: energy identity $\Rightarrow L_t^\infty L_x^2 \cap L_t^2 H_x^1$ bounds $\Rightarrow$ Aubin–Lions compactness $\Rightarrow$ a strong- L^2 subsequential limit which is Leray–Hopf $\Rightarrow$ under U the limit lies in $L_t^\infty H_x^1 \hookrightarrow L_t^\infty L_x^6 \Rightarrow$ the Serrin class $2/s + 3/q = 1/2 \leq 1 \Rightarrow$ smooth [3].*

The central honesty statement — please read before any other claim in this report

The truncated system (2) is the classical Leray/Fourier–Galerkin mollification. That mollified systems are globally smooth, and that a uniform-in-cutoff enstrophy bound implies regularity of the limit, is **classical and standard**. **Hypothesis U is a restatement of the open core of the Millennium problem, not a reduction of its difficulty.** Nothing has been made easier. Its value is purely organisational: one clean quantitative target, four independent attacks, and machine-checked bookkeeping. An earlier version of this programme’s specification overstated this point, and the correction is recorded in docs/REVIEW-2026-08-12.md (finding M1).

3 Dual-Scale Geometry and T-Duality

3.1 Physical motivation (**Tier C**) and its strict boundary

The programme’s geometric inspiration is T-duality in string theory. For a dimension compactified on a circle of radius R , the free-boson partition function is invariant under $R/\sqrt{\alpha'} \rightarrow \sqrt{\alpha'}/R$ with simultaneous exchange of momentum and winding quantum numbers — this is a genuine theorem about the CFT spectrum, established as Eq. (2.2.11) of [14]. The fixed point of that map is $R = \sqrt{\alpha'}$.

Boundary of the analogy — Tier C, quarantined

Reference [14] was retrieved and read for this report, and the distinction it forces is worth stating precisely, because the programme’s earlier internal documents blurred it. The **rigorous** content is the exact $R \leftrightarrow \alpha'/R$ symmetry of the spectrum. The frequently quoted claim that T-duality entails a “**minimal length**” is a *physical interpretation* of that symmetry — the argument being that a string cannot distinguish R from α'/R , so sub- $\sqrt{\alpha'}$ geometry is not resolvable — and is **not itself a separately proved theorem** in the sources consulted.

This report therefore uses T-duality **only** as motivation for the *shape* of a cutoff function. The mechanism acting on the PDE is the Fourier cutoff $J_{\sqrt{\alpha'}}$ and nothing more. Deriving the truncated dynamics from a genuine metric deformation is an *open problem* of this programme (OP-1), not an assumption. All cosmological interpretation — bounce scenarios, dark-sector mass laws, K3/mock-modular numerology — is quarantined in docs/narrative/, may never be imported by lean_src/, and may never be cited in a tier promotion.

3.2 The effective radius: what is actually proven (Tier A)

Define, for $\alpha, R > 0$,

$$R_{\text{eff}}(\alpha, R) := \max\left(R, \frac{\alpha}{R}\right). \quad (4)$$

The following are kernel-verified in `lean_src/CallensDualScale.lean`, with axiom footprint exactly `[propext, Classical.choice, Quot.sound]`:

Formal name	Statement	Reading
<code>Reff_pos</code>	$0 < R_{\text{eff}}(\alpha, R)$	positivity
<code>Reff_ge_sqrt</code>	$\sqrt{\alpha} \leq R_{\text{eff}}(\alpha, R)$	universal minimum scale
<code>Reff_bounce</code>	$R < \sqrt{\alpha} \Rightarrow R_{\text{eff}} = \alpha/R$	sub-cutoff scales reflect
<code>Reff_inertial</code>	$\sqrt{\alpha} \leq R \Rightarrow R_{\text{eff}} = R$	macroscopic scales unmodified
<code>Reff_tdual</code>	$R_{\text{eff}}(\alpha, \alpha/R) = R_{\text{eff}}(\alpha, R)$	self-dual symmetry
<code>Reff_gt_sqrt_of_ne</code>	$R \neq \sqrt{\alpha} \Rightarrow R_{\text{eff}} > \sqrt{\alpha}$	strictness
<code>Reff_eq_sqrt_iff</code>	$R_{\text{eff}} = \sqrt{\alpha} \iff R = \sqrt{\alpha}$	sharpness : the minimum is attained at exactly one point

Scope — what these theorems are and are not

These are theorems about the real-valued function $\max(R, \alpha/R)$. They are elementary, they are correct, and they are *not yet about fluids*. Their role is to fix the geometric language precisely; the parameter α' is carried as a *hypothesis parameter* ($0 < \alpha$), never as an axiom, because Hypothesis U itself quantifies over it. A prior version of this repository asserted these as proven when no such proofs existed; that defect and its correction are recorded in `LEDGER.md` under “Retired / corrected claims”.

3.3 The Symmetric-Square Lock (Tier A)

The programme’s second structural ingredient is an algebraic rigidity constraint. If a sequence satisfies a second-order recurrence with operator L_2 ,

$$u_{n+2} = a u_{n+1} + b u_n,$$

then $v_n := u_n^2$ satisfies the third-order recurrence

$$v_{n+3} = (a^2 + b) v_{n+2} + b(a^2 + b) v_{n+1} - b^3 v_n, \quad (5)$$

which is the symmetric square $L_3 = \text{Sym}^2(L_2)$. If L_2 has characteristic roots $\{\lambda, \mu\}$, then L_3 has roots $\{\lambda^2, \lambda\mu, \mu^2\}$, and the coefficients of (5) are precisely their elementary symmetric functions:

$$e_1 = \lambda^2 + \lambda\mu + \mu^2 = a^2 + b, \quad e_2 = -b(a^2 + b), \quad e_3 = (\lambda\mu)^3 = -b^3.$$

This spectral content is verified as `sym2_symmetric_functions` — it is what makes the identity *structural* rather than a coincidence. The symmetric square of a second-order operator being third-order is classical in the theory of linear differential operators and Picard–Fuchs equations [15].

Intended role (Tier C). The *conjectural* purpose of the lock is to tie macroscopic transport (L_3) rigidly to a microscopic fiber operator (L_2), so that high-frequency modes cannot evolve as independent degrees of freedom — the hoped-for mechanism against Tao-type “choreographed” blow-up [4]. **Its relevance to Navier–Stokes is a conjecture of this programme and is not established.** It must be earned in the dyadic laboratory (§5).

4 The “Holographic Algebra” Organising Concept

Tier and status

The name *HoloAlg* (Holographic Algebra) denotes the programme’s organising picture, not a proven theory. **Everything in this section is Tier C** except where it explicitly cites a Tier A result from §3. It is included because a reader must understand the motivating picture to follow why the four tracks (§7) were chosen — not because the picture carries evidential weight.

4.1 The picture

The organising idea combines two elements already made precise:

1. **A dual-scale geometry** in which the effective scale seen by the fluid is $R_{\text{eff}}(\alpha', R) = \max(R, \alpha'/R)$, so that (i) macroscopic scales are unmodified (`Reff_inertial`: “inertial invisibility”), and (ii) no scale below $\sqrt{\alpha'}$ is ever resolved (`Reff_ge_sqrt`). The regularisation is thereby confined strictly to the sub-cutoff regime.
2. **An algebraic lock** (§3) constraining macroscopic modes to be generated by products of microscopic ones, restricting the admissible spectrum to $\{\lambda^2, \lambda\mu, \mu^2\}$.

The label “holographic” is used in the loose sense that macroscopic (bulk) transport is determined by — and rigidly constrained by — data on a lower-dimensional algebraic structure (the fiber operator’s spectrum). **No claim is made of a relationship to the AdS/CFT correspondence or any holographic duality in the technical sense of theoretical physics.** No such correspondence has been constructed here, and none is used.

4.2 Why this picture would matter, *if* it could be substantiated

For fluid mechanics, the mechanism would matter because it addresses obstruction O1 (Table 1) with a *structural* feature rather than an energetic one: Tao’s averaged system satisfies the energy identity yet blows up, so any successful strategy must use something beyond energy. Algebraic rigidity of the mode spectrum is such a candidate. For astrophysics and cosmology, the same cutoff geometry has been *speculatively* associated with bounce scenarios — but that association is quarantined narrative, carries no evidential weight here, and is deliberately excluded from the mathematical core.

The programme’s own structural finding against itself

When the four tracks’ missing definitions were drafted (`docs/designs/TRACK_DEFINITIONS_DRAFT.md`), the drafting process surfaced a negative structural result about the programme’s own architecture: under the natural choice of entropy functional, **Track T3 collapses into the direct production-identity attack rather than being independent**, and **T1 and T4 both depend on the same unproven embedding (OP-2)**, so agreement between them would be correlated rather than independent evidence. The advertised “four independent attacks” is therefore *not* currently four independent attacks. This finding is recorded rather than suppressed.

5 The Dyadic Laboratory: Established Results

Before touching the continuum PDE, the programme works in a truncated dyadic shell model — a standard toy model of turbulent cascade, introduced by Katz and Pavlović [7], whose published

form (verified by retrieval, as restated in [8]) is $\frac{d}{dt}u_n + \nu 2^{2\alpha n}u_n - \lambda^n u_{n-1}^2 + \lambda^{n+1}u_n u_{n+1} = g_n$. The system used here is the unforced case with λ^n playing the role of k_n : shells $n = 0, \dots, N$ carry wavenumbers $k_n = 2^n$ and real amplitudes a_n , with

$$\frac{da_n}{dt} = \text{NL}_n - \nu k_n^2 a_n, \quad \text{NL}_n := k_{n-1} a_{n-1}^2 - k_n a_n a_{n+1}, \quad (6)$$

with truncation conventions $a_{-1} = 0$, $a_{N+1} = 0$. Energy is $E = \frac{1}{2} \sum a_n^2$ and enstrophy $\Omega = \frac{1}{2} \sum k_n^2 a_n^2$.

Scope note (honesty clause)

This model is a *laboratory*, not the Navier–Stokes equations. Its nonlinearity is **never** asserted in this repository to be derived from, or to approximate, $(u \cdot \nabla)u$. Results below are exact statements about finite sums and their consequences; no existence theory for the ODE system is invoked.

5.1 Energy flux telescoping and monotonicity (Tier A)

In `lean_src/DyadicShells.lean`: the nonlinear energy flux telescopes for *arbitrary* k and a (`dyadic_flux_telescopes`); it vanishes under the truncation $a_{N+1} = 0$ (`dyadic_flux_zero_of_boundary`); and consequently $dE/dt \leq 0$ for $\nu \geq 0$ (`energyRate_nonpos`). Non-vacuity is witnessed: a concrete instance pins $dE/dt = -1$ exactly, so the inequality is not the trivial $0 \leq 0$.

5.2 The exact enstrophy-production identity (Tier A)

The programme’s most substantive dyadic result concerns *enstrophy*, the quantity Hypothesis U is about. Unlike energy flux, the enstrophy production sum does *not* telescope for arbitrary k ; it collapses only under the doubling hypothesis $k_{n+1} = 2k_n$, and it collapses not to zero but to a multiple:

Theorem 5.1 (Dyadic enstrophy production; `enstrophy_production_dyadic`). *Under $k_{n+1} = 2k_n$ and truncation $a_{N+1} = 0$,*

$$\sum_{n=0}^N k_n^2 a_n \text{NL}_n = 3 \sum_{n < N} k_n^3 a_n^2 a_{n+1}, \quad (7)$$

so that the enstrophy evolves as

$$\frac{d\Omega}{dt} = 3 \sum_n k_n^3 a_n^2 a_{n+1} - \nu \sum_n k_n^4 a_n^2. \quad (8)$$

This is the dyadic analogue of vortex stretching, made exact: a signed production term competing against dissipation. The coefficient 3 is not fitted — it is $r^2 - 1$ for doubling ratio $r = 2$, and the Tier B harness independently confirms the general formula at $r = 3$ (coefficient 8). A kernel-verified negative control (`negative_control_nondoubling`) establishes that the identity genuinely *fails* for non-doubling k ; the checker can fail.

5.3 A local production bound (Tier A)

Writing $S_N := \sum_{n < N} k_n^3 a_n^2 a_{n+1}$ and $\Omega_N := \frac{1}{2} \sum_{n \leq N} k_n^2 a_n^2$, `lean_src/EnstrophyProductionBound.lean` proves, in a sqrt-free three-step chain (elementary algebra, $\sum x^2 \leq (\sum x)^2$ for nonnegative x , Cauchy–Schwarz):

$$S_N^2 \leq 2 \Omega_N^3. \quad (9)$$

Precise scope of (§9) — this is where over-reading would be easy

This is a *local, instantaneous, algebraic* bound. It does *not* use dissipation, it does *not* integrate in time, and it does *not* address uniformity in N . Note the exponent: production is bounded by $\Omega^{3/2}$, which is exactly the supercritical scaling that makes the enstrophy differential inequality $\dot{\Omega} \lesssim \Omega^{3/2}$ insufficient for global control. **Inequality (9) is therefore consistent with blow-up and is not evidence for Hypothesis U.** It is infrastructure: an exact, reusable, verified inequality, nothing more. The Tier B mirror further found, honestly, that the negative control breaks Step 1 under non-doubling k but does *not* break the final bound on the tested state family — an unforced negative result, recorded.

6 Statement-Level Formalisation and the Conditional Reduction

6.1 Hypothesis U as a formal statement (Tier A, draft)

`lean_src/HypothesisU_Statements.lean` gives a non-vacuous Galerkin-shape formalisation. Writing $u : \mathbb{R} \rightarrow (\mathbb{N} \rightarrow \mathbb{R})$ for mode amplitudes, a solution at cutoff N satisfies: (i) $u(0) = \text{truncate } N u_0$; (ii) for every t and every retained mode $n \leq N$, $\frac{d}{ds} u(s, n)|_{s=t} = B_n(u(t)) - \nu w_n u(t, n)$; (iii) $u(t, n) = 0$ for $n > N$. Then

$$\text{HypothesisU} \equiv \exists C > 0. \forall N. \forall u. [u \text{ solves at cutoff } N] \Rightarrow \forall t \in [0, T]. \Omega_N(u, t) \leq C. \quad (10)$$

Why the quantifier order is the entire content

$\exists C \forall N$ — the constant is chosen *before* the cutoff. Swapping to $\forall N \exists C$ would be the trivial statement that each finite-dimensional truncation is bounded on its own, which carries no information about the $\alpha' \rightarrow 0$ limit. That swap is the failure mode this ordering exists to exclude.

Two prior defects were repaired by explicit human decision (recorded as Q1, Q2): initial data must be *projected* per cutoff, else the hypothesis set is empty at low N and the statement holds vacuously; and the weight is fixed concretely to $w_n = (2^n)^2$, so that “enstrophy” is provably a nonnegative sum of squares rather than merely a name. The file also *machine-checks the refutation* of the prior formalisation: quantifying over all smooth fields, without the equation as a constraint, yields a statement that is **provably false** (`unconstrained_bound_false`), since scaling a field by c scales enstrophy by c^2 .

The decisive open item

The mode-interaction term B remains an **abstract parameter**. No claim about actual Navier–Stokes solutions may be drawn from this file. Instantiating B with the true 3D nonlinearity is where the entire mathematical content resides, and it is deferred (§9).

6.2 The conditional Millennium reduction (Tier A, draft)

`lean_src/MillenniumReduction.lean` encodes the logical shape of Proposition 5.1 with the two undischarged analytic steps as *named hypothesis parameters*, never axioms:

`millennium_reduction : ( $\forall T$ , HypothesisU)  $\rightarrow$  AubinLionsStatement  $\rightarrow$  ProdiSerrinStatement  $\rightarrow$  GlobalRegu`

The proof is the one-line composition `hPS (hAL hU)`. This is deliberate: all mathematical weight is parked in the hypotheses’ *types*, so the Lean kernel tracks the outstanding debt in the type signature rather than in the axiom environment — the pattern prescribed after a prior version of this work carried a raw `axiom aubin_lions_compactness`.

An adversarial self-review against `SPEC.md`'s actual text found three statement-adequacy gaps, all repaired by explicit decision (Q3–Q5): global regularity had covered only *time*-regularity, and now additionally requires spatial smoothness via membership in every Fourier-side Sobolev class H^s ; the theorem had fixed a single horizon T while concluding a T -independent result, and now quantifies over all T ; and the weight nonnegativity $w \geq 0$ is now an explicit hypothesis.

Audit status — both files are draft

Both files compile clean with correct axiom footprints. **Neither has passed human statement-adequacy audit.** Fixing three found gaps does not certify that no others remain; a subsequent independent review raised two further candidate issues (whether the untruncated solution may differ per horizon T without a coherence clause, and whether the spatial smoothness bound should be uniform in t). Under this programme's charter, authorship — including self-review — never licenses a claim. Only human audit does.

7 The Four Analytical Tracks: Plan and Blocking Status

Stage 3 of the programme proposes to prove Hypothesis U *scale-by-scale* rather than as a monolith, partitioning Fourier space into dyadic shells and controlling $\sum_j E_j^{(\alpha')}(t)$. Four deep analytical disciplines are proposed as independent attacks. Each is an *analogy with a falsification plan*, and each carries a kill criterion agreed in advance.

	Discipline	Proposed mechanism	First falsifiable milestone / kill criterion
T1	Bourgain–Demeter ℓ^2 decoupling [10]	Arithmetic depletion: the Sym^2 constraint starves triadic resonances $k_1 + k_2 = k_3$	Exact count of constrained vs. unconstrained resonant triads. Kill: constrained count grows at the same order $\Rightarrow$ no depletion.
T2	Mouhot–Villani nonlinear Landau damping [11]	Phase mixing suppresses “enstrophy echoes” via a Newton-scheme argument in a high-regularity class	A precise <i>definition</i> of an enstrophy echo in the truncated system, plus its exact evolution. Kill: echoes not suppressed even in the dyadic model.
T3	Golse–Saint-Raymond hydrodynamic limits [12]	A relative-entropy functional controls dissipation uniformly in α'	A relative-entropy inequality for the truncated system at fixed α' . Kill: inequality already fails at fixed α' .
T4	Duminil-Copin percolation [13]	High-enstrophy regions form <i>subcritical</i> percolation clusters $\Rightarrow$ singular set of dimension 0	Recover the CKN dimension bound [5] in the lattice formulation. Kill: cannot recover even CKN.

A near-miss worth recording: the correction that would itself have been an error

An earlier draft of this report flagged an apparent defect in the programme’s specification. The spec describes T2 as resting on *Gevrey-class* regularity; a first retrieval pass indicated that Mouhot–Villani’s *core* result is set in *analytic* regularity, and the obvious move was to “fix” the spec by substituting one for the other.

That fix would have been wrong. Re-fetching the abstract directly — rather than relying on the intermediate summary the first pass produced, parts of which were flagged as paraphrase — gives *both*: the paper establishes “exponential Landau damping in **analytic** regularity”, while the version note records that the main result “now covers Coulomb and Newton potentials, and (2) some classes of **Gevrey** data”. The base theorem is analytic; the final version does cover Gevrey. **The spec was defensible as written and was left standing.**

The methodological point is the reason this is in the report at all: the first pass had enough material to justify editing the normative rulebook, and editing it would have *introduced* an error on the strength of a paraphrase — exactly the failure mode rule LL-6 exists to prevent, one level up from where it was first learned. **When a proposed correction rests on a summary rather than a primary quotation, flag it and re-fetch; do not edit the rulebook.**

One substantive caveat did survive, and it is the part that bears on feasibility: coverage is “*some* classes of Gevrey data”, not all. Any T2 argument leaning on Gevrey regularity must identify *which* class and must not assume the full Gevrey scale.

A disanalogy that has not been resolved away

The Golse–Saint-Raymond programme [12] concerns the Boltzmann-to-Navier–Stokes hydrodynamic limit, whereas this programme’s $\alpha' \rightarrow 0$ limit is **not** a Knudsen limit and has no kinetic layer — a disanalogy flagged internally (review finding M4) and repeated here so T3 is not over-read. This one stands.

7.1 Current status: correctly blocked

All four tracks are blocked on undefined objects — and this is the correct state

Tracks T1–T4 require four objects that **do not exist** in this programme or, to the authors’ knowledge, in the literature:

- **OP-2**: the Sym^2 -constrained spectrum on $\mathbb{T}^3$. The lock is proven for *scalar recurrences*; its embedding into 3D Fourier dynamics indexed by $k \in \mathbb{Z}^3$ is open. Any invented embedding would make T1’s count meaningless.
- **OP-3**: “enstrophy echo” as a formula. Landau-damping echoes are defined in Vlasov phase space; no NSE analogue exists.
- **OP-4**: the entropy functional h and reference state $\bar{u}$. Choosing h is the mathematical content of T3.
- **OP-5**: the coupling of the percolation field to Sym^2 . Without it, T4 measures generic percolation, which cannot beat CKN.

Under escalation rule E-1, an agent encountering a missing definition must **stop and escalate**, never improvise. Draft definitions have been authored and await human audit; *authorship never unblocks a track — only audit does*. Progress here is deliberately gated on human judgement rather than machine throughput.

7.2 Executable instrumentation delivered (Tier B)

Definition-independent tooling has nonetheless been built and certified, so that the tracks can execute immediately once unblocked: exact lattice representation counts $r_3(n)$ for $n \leq 10^4$ with Legendre anchors; exact unconstrained triad counts for $|k_i|^2 \leq M^2$; and an exact 3D periodic site-percolation toolkit (union-find over integers, wrap detection, 27 checks, negative control that fails when periodicity is dropped).

8 Numerical Investigation and Its Two Failures

The central empirical question is whether $\sup_t \Omega_N(t)$ grows with the cutoff N or saturates. Two successive designs *failed their feasibility premises*, for genuinely different reasons; both are documented rather than quietly replaced.

1. **Explicit float RK4 (D4).** Stiffness: stability requires $\Delta t \sim 1/(\nu k_N^2)$, which collapses as N grows. 21 of 36 configurations were computationally infeasible.
2. **Exact-rational IMEX-Euler (D5).** The implicit viscous treatment removed the stiffness barrier as designed — but the scheme failed anyway, for an unrelated reason: iterating a *nonlinear* rational map in exact arithmetic causes the numerators and denominators to grow geometrically in bit-length (hundreds of digits within 2–6 steps), while the state magnitude stays small. **This is a representation blow-up, not a numerical instability.** Implicit-versus-explicit was the wrong axis on which to redesign.

Generalised lesson, now a standing rule

Exact rational arithmetic is excellent for closed-form algebraic identities and inequalities proven in one shot, and poor for *iterating* a nonlinear map over many steps. For trajectories, use dual-precision floating point (Tier C steering) and reserve exactness for the algebra. After two failed designs, the methodology decision was escalated to the human owner rather than a third variant being chosen unilaterally.

8.1 Resolution and the current evidence base (Tier C steering data)

The accepted resolution was to stop pursuing exact-rational certification and retain dual-precision (float64 + 50-digit `mpmath`) runs as permanent Tier C steering data. Across the full grid $N \in \{8, 12, 16, 20, 24\}$, several low-viscosity configurations initially reported divergence, with the two precisions agreeing to $\sim 10^{-14}$ — ruling out rounding, but *not* distinguishing genuine divergence from an under-resolved step size (the design's Δt formula never depended on ν). A step-size refinement study resolved this:

Finding	Detail
Divergences are artifacts	All four diverging (ν , profile) pairs flip to OK at a finite refinement level and stay OK at every finer level, with $\sup \Omega$ <i>decreasing</i> as $\Delta t \rightarrow 0$ (e.g. $45.1 \rightarrow 21.8 \rightarrow 17.3 \rightarrow 15.8$) — the signature of a discretisation artifact resolving, not a fixed-time blow-up.
N -independence	44 of 45 configurations are <i>exactly</i> flat in N (bit-identical $\sup \Omega$ at every cutoff).
The one exception, explained	$\nu = 0.1$, profile P2 appeared to grow linearly in N (4.5, 6.5, $\dots$, 12.5). Traced by hand: this equals $\frac{1}{2}(N + 1)$, which is profile P2's <i>analytic initial</i> enstrophy — P2 excites every retained mode by construction, so a larger cutoff starts with more enstrophy independent of any dynamics. The trajectory collapses immediately from $t = 0$ and decays thereafter. An initial-data artifact, not dynamical growth.

No verdict is drawn

Once the P2 case is understood, every tested configuration is flat in N . Under PLAN.md §8, a verdict on uniformity is reserved to the human owner and has **not been issued**. These are Tier C floating-point measurements of a *specific discrete scheme* — not of the continuum dyadic model, and certainly not of Navier–Stokes.

9 Toward Instantiating the Nonlinearity (OP-6)

Because B in (10) is abstract, the programme's decisive next step is to instantiate it. A repository-wide search established that **no Fourier-triad convolution formula exists anywhere in this programme's sanctioned content** — the triad tooling counts lattice points without amplitudes; the OP-2 draft self-admits it discards the angular structure of k “which is precisely where 3-D vortex-stretching geometry lives”; and the dyadic NL_n is never asserted to derive from $(u \cdot \nabla)u$. Instantiating B is therefore a genuine E-1 blocker.

On $\mathbb{T}^3$, with $u = \sum_k \hat{u}_k e^{ik \cdot x}$, $k \cdot \hat{u}_k = 0$, $\hat{u}_{-k} = \overline{\hat{u}_k}$, and Leray projector $P(k) = I - \frac{k \otimes k}{|k|^2}$, the truncated nonlinearity is

$$\partial_t \hat{u}_k = -\nu |k|^2 \hat{u}_k - i \sum_{\substack{p+q=k \\ p,q \in \Lambda}} P(k) [(q \cdot \hat{u}_p) \hat{u}_q]. \quad (11)$$

An erratum caught by exact arithmetic — a case study in why the method exists

The formula first recorded in the scoping memo, taken from a single literature cross-check, read $P(k)[q(\hat{u}_p \cdot \hat{u}_q)]$ — the *vector* q times the scalar velocity–velocity dot product. The Tier B harness then *proved*, in exact arithmetic, that this expression is **identically zero for every k** : $P(k)$ annihilates anything parallel to k , and since every pair (p, q) with $p + q = k$ appears alongside its swap (q, p) , the two terms cancel exactly, $P(k)[qs] + P(k)[ps] = P(k)[(p + q)s] = P(k)[ks] = 0$. That cannot be the real nonlinearity. The corrected form (11) — q *dotted* with $\hat{u}_p$, a scalar, times the vector $\hat{u}_q$ — matches the Fourier transform of the convective derivative and is nonzero. The error was caught by computation, not by inspection.

The resulting harness (tests/tier_b_nse_triad_convolution.py, **Tier B**) certifies two structural facts in exact Gaussian-rational arithmetic at $M = 1, 2, 3$: *transversality* $k \cdot N(u)_k = 0$

(unconditional, since $P(k)$ projects onto $k^\perp$), and *detailed energy conservation* $\sum_k \overline{\hat{u}_k} \cdot N(u)_k = 0$ given divergence-free, conjugate-symmetric input. Two negative controls — dropping the Leray projection, and breaking the divergence-free condition on one mode — each break exactly one fact and leave the other intact, matching the hand derivation rather than merely “changing something”.

Recorded honestly: several negative controls were undetectable by construction

Two natural “plausible bug” perturbations turned out to be invisible: swapping $p \leftrightarrow q$ throughout (the pair set is symmetric under relabelling for *every* fixed k , not merely after summing), and substituting the outer k for the summed q (the difference vanishes because $p \cdot \hat{u}_p = 0$ by divergence-freeness). The negative control actually used — multiplying by the wrong velocity factor — was found by *testing* candidates, not by assuming one would work. Additionally, the general triad identity underlying energy conservation is here verified *computationally* at three truncations, **not proven symbolically**; a Tier A proof remains future work.

9.1 From computational check to kernel proof (Tier A)

The harness above certifies two facts *computationally*, at three truncations, and its own docstring declines to claim either as proven. Both are now Tier A, in `lean_src/TriadConservation.lean`:

Theorem 9.1 (Termwise triad cancellation; `triad_pairing`). *If $k_p + k_q + k_r = 0$ componentwise and the field is divergence-free at p ($k_p \cdot u_p = 0$), then the two orderings of the triad summand cancel exactly:*

$$(k_q \cdot u_p)(u_q \cdot u_r) + (k_r \cdot u_p)(u_r \cdot u_q) = 0.$$

Why the first attempt failed, and what actually worked

The harness records that a direct *three-way* relabelling (cycling $p \rightarrow q \rightarrow r$) did not close. It does not need to. The cancellation is **two-way**, and stronger than a summation argument: it is **termwise**. Pairing (p, q, r) with (p, r, q) — swapping only the last two indices, leaving p fixed — and using symmetry of the pairing followed by additivity,

$$[(k_q \cdot u_p) + (k_r \cdot u_p)](u_q \cdot u_r) = ((k_q + k_r) \cdot u_p)(u_q \cdot u_r) = (-k_p \cdot u_p)(u_q \cdot u_r) = 0.$$

This was re-derived by hand and confirmed on 6486 triples in exact arithmetic *before* any Lean was written. Summing it over any finite index set closed under the swap gives detailed energy conservation (`triad_sum_zero`); transversality (`transversality_of_sum`) is separate and unconditional, following from the projector landing in $k^\perp$.

Scope: this does not close OP-6

The file is stated over an *arbitrary commutative ring*, with an arbitrary additive index group and additive wavevector map. It is deliberately **not** stated over $\Lambda \subset \mathbb{Z}^3$ with complex velocities, because that concrete apparatus *is* the open decision D1. **The bridge from this lemma to the concrete Fourier–Galerkin setting is not built**; nothing here instantiates B ; nothing here is a statement about Navier–Stokes solutions. One hypothesis is load-bearing rather than technical: the swap has fixed points (where $q = r$, forcing $p = -2q$) at which the pairing yields only $2f = 0$, so the ring must have no 2-torsion. For the intended instantiations $(\mathbb{R}, \mathbb{C})$ this holds.

Open decisions. Instantiating B is not a substitution but a restructuring: the state representation must move from $\mathbb{N} \rightarrow \mathbb{R}$ shell scalars to $\mathbb{Z}^3 \rightarrow \mathbb{C}^3$ Fourier vectors carrying divergence-free and reality constraints. Whether to undertake that full reindexing or seek a smaller shell-preserving proxy, and whether to sequence it before or after OP-2, are open questions reserved to the human owner.

10 What Has Failed, and What That Taught

A programme that reports only successes is not reporting. The following incidents are recorded in `LL.md` with evidence, and each produced a standing process rule.

ID	Incident	Resulting rule
LL-1	A broad <code>git add -A</code> swept an agent’s mid-edit Lean file into a commit; the committed version carried <code>sorryAx</code> in five theorems. Happened <i>twice</i> in one session.	Stage by explicit path only while any background process may be writing; re-run the gate on the exact file immediately before staging.
LL-2	A submitted proposal self-reported “proven, no <code>sorry</code> ”; it did not compile (four Lean idiom errors) and its footprint contained <code>sorryAx</code> .	Never trust a self-report. Re-run the compiler on the exact artifact. Archive proposals verbatim with their compiled kernel output.
LL-3	D5’s redesign fixed stiffness, but the actual obstruction was representation-size growth.	Diagnose the failure mode before redesigning; a fix for the wrong axis is no fix.
LL-4	Two successive designs by the same author failed their feasibility premises.	A repeated failed design is a signal to <i>escalate the methodology decision</i> , not to try a third variant alone.
LL-6	A literature claim was nearly cited from memory; the fetched abstract showed the real result was narrower (well-posedness of <i>positive</i> solutions in a particular scaling range, not unconditional regularity) [9].	Verify literature precisely before citing, even for a well-known author or result.
LL-7	A top-tier-authored design memo’s own worked example quoted the wrong summation range.	The certifying agent must trace and correct rather than silently absorb; “derive before dispatch” does not remove the need for downstream re-verification.
LL-8	The verification gate itself had a false positive. A lemma proved by purely computational means reports the <i>strictly cleaner</i> footprint [<code>propext</code>]; the gate string-matched the exact three-axiom list and rejected it as unsound.	Test a changed gate predicate against a table including cases that <i>must</i> fail, before trusting a real run; and confirm the script’s exit code , not its trailing output. The rule’s wording in <code>SPEC.md</code> was amended to match its intent (membership, not equality).
—	The Mathlib dependency was never pinned. The <code>lakefile</code> said <code>require mathlib from git</code> with no revision; a cold build would have resolved to master and verified everything against a different Mathlib than all existing claims were checked with.	Pin the verifier itself by revision, and track the lockfile. An unpinned prover is a defect in the foundation of a machine-checked programme, not a build-script nuisance.

ID	Incident	Resulting rule
—	A proposed correction to the spec would itself have been an error (the T2 regularity class, §7). The evidence for it was a paraphrased summary, not a primary quotation.	When a correction rests on a summary rather than a direct quote, flag and re-fetch — do not edit the normative rulebook. LL-6 applied to its own application.

11 Summary of Verified State

Artifact	Content	Theorems
CallensDualScale.lean	T-dual R_{eff} laws, Sym^2 lock, sharpness	14
DyadicShells.lean	Energy-flux telescoping, monotonicity	3
HypothesisU_Statements.lean	Hypothesis U statement shape (DRAFT)	9
EnstrophyProduction.lean	Exact enstrophy-production identity	11
EnstrophyProductionBound.lean	Local bound $S_N^2 \leq 2\Omega_N^3$	15
MillenniumReduction.lean	Conditional reduction skeleton (DRAFT)	7
TriadConservation.lean	Detailed energy conservation + transversality	7
Total, no axiom outside {propext, Classical.choice, Quot.sound}		66

Seven Tier B harnesses are wired into Gate 1, each with a negative control demonstrated to fail. Both gates pass at commit 36c4bae.

11.1 What is genuinely established

Exact algebraic identities and inequalities for a dyadic shell model; the T-dual effective radius laws with sharpness; a non-vacuous statement-level formalisation of Hypothesis U including a machine-checked refutation of a prior, provably false formulation; a conditional reduction skeleton with the analytic debt tracked in types; and exact-arithmetic verification of two structural properties of the Fourier–Galerkin nonlinearity.

11.2 What is emphatically not established

Hypothesis U itself; any statement about actual Navier–Stokes solutions; any claim that the Symmetric-Square Lock is relevant to the NSE cascade; the four tracks (all blocked on undefined objects); any uniformity verdict; and the statement-adequacy of the two DRAFT Lean files. The instantiation of B — where, by the programme’s own assessment, “the entire mathematical content resides” — has not been done.

Definition of done, and what is absent from it

The campaign is complete when the formal skeleton is merged and human-audited, the dyadic results are kernel-verified, the datasets carry a human-issued verdict *whatever that verdict is*, the open problems are either audited or recorded as open with dated escalations, and every ledger row maps to a passing artifact. Note what is *absent*: proving Hypothesis U. If this programme produces a negative dyadic verdict or kills tracks, it **still counts as done and successful as science**. That is what falsifiable structure means.

Note on reference verification

Consistent with this programme’s own rule LL-6 (“*verify literature precisely before citing, even a well-known author or result*”), every reference below was verified by retrieving the actual arXiv abstract page, publisher listing, or official PDF at the time of writing. **No citation here is written from memory.** Where a detail could not be confirmed by direct retrieval, it is explicitly marked [UNVERIFIED] rather than being asserted. Readers should treat those specific marked details as requiring independent confirmation before reuse.

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